Supplementary Material S4. Categorized Taxonomy of the Included Studies: Data Underlying Table III (n = 71)

Categories correspond to Table III of the paper. "No." refers to the record number in S1; "Ref." refers to the reference number in the paper.

Year = journal volume year (year of online publication for articles not yet assigned to a volume). Best-performing model as reported by the original authors.

Database	No.	Ref.	Study	Year	Application domain	Prediction target	Modeling paradigm	Best-performing model	Best-model group	XAI techniques	No. of XAI techniques	Explanation level	Interaction effects	Counterfactual explanations	Explanation reliability assessment	Performance metrics
Scopus	72	8	Farooq et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML only	ETR	Tree-based ensemble	LIME	1	Local only	No	No	Not examined	R^2, RMSE, MAE, MSE
Scopus	70	9	Ranjan and Madanayak (2026)	2026	Transportation	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP, PDP	2	Global + Local	No	No	Not examined	R^2, RMSE, MAE, MSE
Scopus	110	10	Şahin (2025)	2025	Transportation	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global only	No	No	Not examined	R^2, RMSE, MAE, Relative MAE (%)
Scopus	42	11	Özüpak (2026)	2026	Transportation	Carbon/GHG emissions	ML and DL	Stacked Ensemble	Ensemble / stacking	SHAP, LIME	2	Global + Local	No	No	Not examined	RMSE, MAE, R^2, AUC, NSE, Theil U2
Scopus	50	12	Gungor et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML and DL	ANN	Deep learning	SHAP, Permutation / feature importance	2	Global + Local	No	No	Qualitative cross-method consistency	RMSE, MSE, MAE, MedAE, Bias, R^2, MAPE, NRMSE, EVS, Corr, RMSLE
Scopus	32	13	Wani et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML and DL	CatBoost (CB)	Tree-based ensemble	SHAP, LIME, PDP	3	Global + Local	No	No	Quantitative cross-method agreement	R^2, MAE, RMSE, MSE, NSE, MAPE
Scopus	16	14	Zhu et al. (2026)	2026	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP, PDP	2	Global + Local	Yes	No	Not examined	R^2, MSE, MAE
Scopus	23	15	Song et al. (2026)	2026	Agriculture	Carbon/GHG emissions	ML and DL	XGB	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, RMSE, MAE
Scopus	29	16	Song et al. (2026)	2026	Energy and power systems	Carbon/GHG emissions	ML and DL	MOIRAI-2 + UPINN	Deep learning	Attention-based, Counterfactual (DiCE etc.), Intrinsic / rule-based / physics	3	Global + Local	No	Yes	Not examined	MAE, RMSE, R^2
Scopus	192	17	Assael et al. (2023)	2023	Corporate carbon accounting	Carbon/GHG emissions	ML only	LightGBM	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, RMSE, MAE
Scopus	108	18	Zhou et al. (2025)	2025	Carbon markets	Carbon/allowance price	ML and DL	DNMADE	Deep learning	SHAP, Attention-based	2	Global + Local	No	No	Not examined	MAE, RMSE, MAPE, MSPE, IA; PICP, PINAW, AIS, CWC; DA
Scopus	158	19	Liang et al. (2025)	2025	Carbon markets	Carbon/allowance price	ML and DL	EEMD–GAMI_NET–LSTM–ARIMA	Deep learning	ALE	1	Global only	No	No	Not examined	R^2, RMSE, MAE, DA
Scopus	4	28	Ye et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML only	CatBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, RMSE, MAE, MAPE
Scopus	2	29	Özüpak and Aslan (2027)	2027	National and global emissions	Carbon/GHG emissions	ML only	Optimized RF	Tree-based ensemble	SHAP, LIME	2	Global + Local	No	No	Qualitative cross-method consistency	R^2, RMSE, MAE, MSE, Max Error
Scopus	41	30	Yang et al. (2026)	2026	Carbon markets	Carbon/allowance price	ML and DL	XCPFNet (CEEMDAN–CNN–LSTM–Transformer)	Deep learning	SHAP	1	Global + Local	Yes	No	Stability	MSE, MAE, RMSE, MAPE, R^2; PIPC, PINAW; ES, FIC, TS
Scopus	1	31	Ali et al. (2027)	2027	Agriculture	Carbon/GHG emissions	ML only	Random Forest Regressor (RF)	Tree-based ensemble	SHAP, LIME, Counterfactual (DiCE etc.)	3	Global + Local	No	Yes	Not examined	R^2, MSE, RMSE, MAE, MAPE, Median Absolute Error, Max Error, Explained Variance Score
Scopus	8	32	Saraswat et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML and DL	GAO-MLP (MLGO)	Deep learning	SHAP, Permutation / feature importance, Grad-CAM / CAM	3	Global + Local	No	No	Not examined	RMSE, R^2, MAPE, MARE, TIC
Scopus	27	33	Shaik et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML and DL	FFO-TabNet	Deep learning	SHAP, LIME, Intrinsic / rule-based / physics	3	Global + Local	No	No	Qualitative cross-method consistency	RMSE, MAE, R^2, MAPE
Scopus	90	34	Saleem et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML only	Extra Trees (FMS-selected)	Tree-based ensemble	SHAP, LIME	2	Global + Local	No	No	Not examined	R^2, RMSE, MAE
Scopus	109	35	Alam et al. (2025)	2025	Transportation	Carbon/GHG emissions	DL only	CarbonMLP (MLP)	Deep learning	SHAP	1	Global + Local	No	No	Not examined	R^2, MSE, RMSE, MAPE

Database	No.	Ref.	Study	Year	Application domain	Prediction target	Modeling paradigm	Best-performing model	Best-model group	XAI techniques	No. of XAI techniques	Explanation level	Interaction effects	Counterfactual explanations	Explanation reliability assessment	Performance metrics
Scopus	131	36	Yuan et al. (2025)	2025	Transportation	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	ALE, Permutation / feature importance	2	Global only	No	No	Not examined	R^2, MSE, RMSE
Scopus	167	37	Guo et al. (2024)	2024	Transportation	Carbon/GHG emissions	ML only	GBM	Tree-based ensemble	SHAP, ALE	2	Global only	No	No	Not examined	R^2, RMSE, rRMSE, MAE, MAPE
Scopus	33	38	Luo et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP, ICE	2	Global + Local	No	No	Not examined	R^2, RMSE
Scopus	28	39	Chen et al. (2026)	2026	Transportation	Carbon/GHG emissions	ML only	ET	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	MSE, RMSE, R^2
Scopus	116	40	Huang et al. (2025)	2025	Transportation	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	RMSE, MAE, R^2
Scopus	133	41	Zhao and He (2025)	2025	Transportation	Carbon/GHG emissions	ML and DL	CatBoost	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, RMSE, MAE, MAPE
Scopus	137	42	Roy et al. (2025)	2025	Transportation	Carbon/GHG emissions	ML and DL	MT2CN	Deep learning	SHAP	1	Global + Local	Yes	No	Not examined	R^2, RMSE, MAE, MAPE
WoS only	70	43	Liu et al. (2025)	2025	Transportation	Carbon/GHG emissions	ML only	CatBoost	Tree-based ensemble	SHAP, PDP	2	Global + Local	No	No	Not examined	R^2, RMSE, MAE
Scopus	73	44	Fang et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML and DL	DCG-TCN-GCN	Deep learning	SHAP	1	Global + Local	No	No	Not examined	Corr, RSE, RMSE
Scopus	114	45	Zhang et al. (2025)	2025	Buildings and construction	Carbon/GHG emissions	ML only	RF	Tree-based ensemble	SHAP	1	Global only	No	No	Not examined	R^2, RMSE
Scopus	193	46	Zhang et al. (2023)	2023	Buildings and construction	Carbon/GHG emissions	ML only	LightGBM	Tree-based ensemble	SHAP, PDP	2	Global + Local	No	No	Not examined	R^2, MSE, RMSE, MAE
WoS only	75	47	Zhou et al. (2024)	2024	Buildings and construction	Carbon/GHG emissions	ML only	LightGBM	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, MAPE, MAE
Scopus	180	48	Khan et al. (2024)	2024	Buildings and construction	Carbon/GHG emissions	ML only	BO-LGBM	Tree-based ensemble	LIME	1	Local only	No	No	Not examined	R^2, MAPE
Scopus	61	49	Cao et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML only	BO-CatBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, RMSE, MAE
Scopus	59	50	Wang et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML and DL	XGB	Tree-based ensemble	SHAP, PDP	2	Global + Local	No	No	Not examined	R^2, MAE, RMSE, MAPE
Scopus	51	51	Seo et al. (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML only	DT + SVR Ensemble	Ensemble / stacking	SHAP	1	Global + Local	Yes	No	Not examined	MAE, SMAPE
Scopus	88	52	Zheng and Shuang (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML and DL	CatBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	MAE, MAPE, RMSE, R^2
Scopus	98	53	Zawaideh (2026)	2026	Buildings and construction	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global only	No	No	Not examined	R^2, RMSE, MAE, MAPE
Scopus	115	54	Liu et al. (2025)	2025	Buildings and construction	Carbon/GHG emissions	ML only	GBRT	Tree-based ensemble	SHAP, PDP, ICE	3	Global + Local	No	No	Not examined	MSE, MAE, RMSE, R^2, CWI
Scopus	17	55	Chen et al. (2026)	2026	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, MSE, RMSE, MAE
Scopus	94	56	Ren et al. (2026)	2026	Urban, regional, and land-use systems	Carbon/GHG emissions	ML and DL	XGBoost	Tree-based ensemble	SHAP, Grad-CAM / CAM	2	Global + Local	Yes	No	Faithfulness	Accuracy, Precision, R^2, MAE, RMSE, EVS
Scopus	95	57	Bhowmik et al. (2026)	2026	Urban, regional, and land-use systems	Carbon/GHG emissions	ML and DL	RF (Tree cover loss), XGBoost (CO_2 emissions)	Tree-based ensemble	SHAP	1	Global only	No	No	Not examined	R^2, RMSE, MAE
Scopus	99	58	Jiang et al. (2026)	2026	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, Kappa
Scopus	130	59	Bilotta et al. (2025)	2025	Urban, regional, and land-use systems	Carbon/GHG emissions	ML and DL	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, MAE, MSE, RMSE, MAPE
Scopus	132	60	Zhang et al. (2025)	2025	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, RMSE, MAE, PRD
Scopus	152	61	Xu et al. (2025)	2025	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, MAE, MSE, RMSE, MAD, MAPE, EVS, MSLE
Scopus	155	62	Wang et al. (2025)	2025	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	LGBM	Tree-based ensemble	SHAP	1	Global + Local	Yes	No	Not examined	R^2, RMSE, MAE
Scopus	185	63	Luo et al. (2024)	2024	Urban, regional, and land-use systems	Carbon/GHG emissions	ML and DL	Optimized ETR	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	R^2, MSE, RMSE, MRE, MAE, MBE

Database	No.	Ref.	Study	Year	Application domain	Prediction target	Modeling paradigm	Best-performing model	Best-model group	XAI techniques	No. of XAI techniques	Explanation level	Interaction effects	Counterfactual explanations	Explanation reliability assessment	Performance metrics
Scopus	191	64	Yao et al. (2023)	2023	Urban, regional, and land-use systems	Carbon/GHG emissions	ML only	GA-EKC (preferred turning-point model)	Other / not established	SHAP	1	Global + Local	No	No	Not examined	R^2, Adjusted R^2
Scopus	5	65	Sireesha et al. (2026)	2026	National and global emissions	Carbon/GHG emissions	ML only	GB, XGBoost	Tree-based ensemble	SHAP, LIME, PDP, ICE	4	Global + Local	No	No	Faithfulness/fidelity; quantitative cross-method agreement	R^2, RMSE, MAE, VAF
Scopus	74	66	Dahmani et al. (2026)	2026	National and global emissions	Carbon/GHG emissions	DL only	MTL-Conv1D-LSTM	Deep learning	LIME	1	Local only	No	No	Not examined	R^2, MAE, MSE, RMSE
Scopus	76	67	Ghannam (2026)	2026	National and global emissions	Carbon/GHG emissions	ML and DL	Random Forest (RF)	Tree-based ensemble	SHAP, Permutation / feature importance	2	Global only	No	No	Stability	R^2, RMSE, MAE, MSE, MAPE, MSLE, MedAE
Scopus	87	68	Ma et al. (2026)	2026	National and global emissions	Carbon/GHG emissions	ML and DL	MLP	Deep learning	SHAP	1	Global + Local	No	No	Not examined	R^2, MSE, RMSE, MAE, MAPE
Scopus	151	69	Zheng and Shuang (2025)	2025	National and global emissions	Carbon/GHG emissions	ML and DL	CatBoost	Tree-based ensemble	SHAP	1	Global + Local	No	No	Not examined	MAE, RMSE, MAPE, R^2
Scopus	153	70	Han and Lin (2025)	2025	National and global emissions	Carbon/GHG emissions	ML and DL	ANN	Deep learning	SHAP	1	Global + Local	No	No	Not examined	MAE, MSE, RMSE, R^2
Scopus	156	71	Chen et al. (2025)	2025	National and global emissions	Carbon/GHG emissions	ML only	CBRB-IC	Other / not established	SHAP, Intrinsic / rule-based / physics	2	Global + Local	No	No	Not examined	MAE, MSE, RMSE, MAPE, R; modeling time, inference time, rule count, antecedent attribute count
Scopus	187	72	Mangla et al. (2024)	2024	National and global emissions	Carbon/GHG emissions	ML and DL	Not explicitly established	Other / not established	SHAP	1	Global + Local	No	No	Qualitative cross-method consistency	R^2, Adjusted R^2 (regression validation); Support, Confidence, Lift (association rules)
Scopus	203	73	Crespo et al. (2021)	2021	National and global emissions	Carbon/GHG emissions	ML and DL	Proposed NN/BP-RReliefF Framework	Deep learning	LIME, Permutation / feature importance	2	Global + Local	No	No	Not examined	RMSE, MAPE
Scopus	54	74	Shi et al. (2026)	2026	Energy and power systems	Carbon/GHG emissions	ML only	XGBoost	Tree-based ensemble	SHAP	1	Global only	Yes	No	Not examined	R^2, RMSE
Scopus	75	75	Qader and Albalooshi (2026)	2026	Energy and power systems	Carbon/GHG emissions	ML and DL	XGBoost	Tree-based ensemble	SHAP, LIME	2	Global + Local	No	No	Not examined	RMSE, MAE, R^2, MAPE
Scopus	78	76	Ramalingam et al. (2026)	2026	Energy and power systems	Carbon/GHG emissions	ML only	GBM	Tree-based ensemble	SHAP, Permutation / feature importance	2	Global + Local	No	No	Qualitative cross-method consistency	R^2, RMSE, MSE, MAE
Scopus	100	77	Sayghe et al. (2026)	2026	Energy and power systems	Carbon/GHG emissions	ML and DL	GridFM	Deep learning	SHAP, Attention-based	2	Global only	No	No	Not examined	MAPE, sMAPE, RMSE, MAE, CRPS
Scopus	71	78	Sireesha & Sheik (2026)	2026	Agriculture	Carbon/GHG emissions	DL only	GRU	Deep learning	SHAP, LIME, PDP, ICE	4	Global + Local	No	No	Quantitative cross-method agreement	R^2, RMSE, MAE, VAF
Scopus	157	79	Jeaunita et al. (2025)	2025	Agriculture	Carbon/GHG emissions	ML and DL	FNN (carbon emission prediction); Deep Autoencoder (anomaly detection)	Deep learning	SHAP, LIME	2	Global + Local	No	No	Not examined	R^2, MSE, RMSE, Accuracy, Precision, Recall, F1 Score
Scopus	45	80	Chai et al. (2026)	2026	Corporate carbon accounting	Carbon/GHG emissions	DL only	Ensemble (Transformer + CNN + BiLSTM)	Deep learning	SHAP, LIME	2	Global + Local	Yes	No	Not examined	MAPE, RMSE, Pearson r, Emission reduction (%), ROI, NPV, Payback period
Scopus	24	81	Li et al. (2026)	2026	Carbon markets	Carbon/allowance price	ML only	RW	Other / not established	SHAP, PDP, Permutation / feature importance	3	Global + Local	No	No	Stability	RMSE, MAE, MAPE, DA
Scopus	25	82	Ma et al. (2026)	2026	Carbon markets	Carbon/allowance price	ML and DL	VC ensemble	Ensemble / stacking	SHAP	1	Global + Local	No	No	Not examined	MSE, MAE, RMSE, MAPE, R^2
Scopus	44	83	Sun et al. (2026)	2026	Carbon markets	Carbon/allowance price	ML and DL	CaStNet	Deep learning	SHAP, LIME	2	Global + Local	Yes	No	Not examined	RMSE, MAE, MAPE, R^2
Scopus	83	84	He et al. (2026)	2026	Carbon markets	Carbon/allowance price	ML and DL	CEEMDAN–CNN–BiGRU–Attention[1]	Deep learning	SHAP, LIME	2	Global + Local	No	No	Qualitative cross-method consistency	MAE, RMSE, MAPE, R^2

Database	No.	Ref.	Study	Year	Application domain	Prediction target	Modeling paradigm	Best-performing model	Best-model group	XAI techniques	No. of XAI techniques	Explanation level	Interaction effects	Counterfactual explanations	Explanation reliability assessment	Performance metrics
Scopus	171	85	Sayed et al. (2024)	2024	Carbon markets	Carbon/allowance price	ML and DL	MLSO–LSTM	Deep learning	SHAP	1	Global + Local	No	No	Not examined	RMSE, R^2, MAE, MAPE, MSE
Scopus	183	86	Deng et al. (2024)	2024	Carbon markets	Carbon/allowance price	ML and DL	CEEMDAN–TFS–BOA–LightGBM	Deep learning	SHAP	1	Global + Local	No	No	Not examined	AUC, Accuracy, Precision, Recall, F1 score

Summary of the Categorized Taxonomy (N = 71)

Studies may use several models or XAI techniques; percentages are relative to N = 71.

Dimension	Category	Studies (n)	Share (%)
Prediction target	Carbon/GHG emissions	62	87.3
Prediction target	Carbon/allowance price	9	12.7
Application domain	Transportation	17	23.9
Application domain	Buildings and construction	13	18.3
Application domain	Urban, regional, and land-use systems	11	15.5
Application domain	National and global emissions	10	14.1
Application domain	Carbon markets	9	12.7
Application domain	Energy and power systems	5	7.0
Application domain	Agriculture	4	5.6
Application domain	Corporate carbon accounting	2	2.8
Modeling paradigm	ML only	34	47.9
Modeling paradigm	ML and DL	33	46.5
Modeling paradigm	DL only	4	5.6
Best-model group	Tree-based ensemble	42	59.2
Best-model group	Deep learning	22	31.0
Best-model group	Other / not established	4	5.6
Best-model group	Ensemble / stacking	3	4.2
XAI technique	SHAP	64	90.1
XAI technique	LIME	17	23.9
XAI technique	PDP	10	14.1
XAI technique	ICE	4	5.6
XAI technique	ALE	3	4.2
XAI technique	Permutation / feature importance	7	9.9
XAI technique	Attention-based	3	4.2
XAI technique	Counterfactual (DiCE etc.)	2	2.8
XAI technique	Grad-CAM / CAM	2	2.8
XAI technique	Intrinsic / rule-based / physics	3	4.2
XAI technique	Single XAI technique only	38	53.5
Explanation level	Global + Local	58	81.7
Explanation level	Global only	10	14.1
Explanation level	Local only	3	4.2
Explanation level	Interaction effects analyzed	14	19.7
Explanation evaluation	Reliability examined	13	18.3
Explanation evaluation	Counterfactual explanations	2	2.8
Explanation evaluation	Causal validity assessed	0	0.0
Publication year	2021	1	1.4
Publication year	2023	3	4.2
Publication year	2024	7	9.9
Publication year	2025	19	26.8
Publication year	2026	39	54.9
Publication year	2027	2	2.8